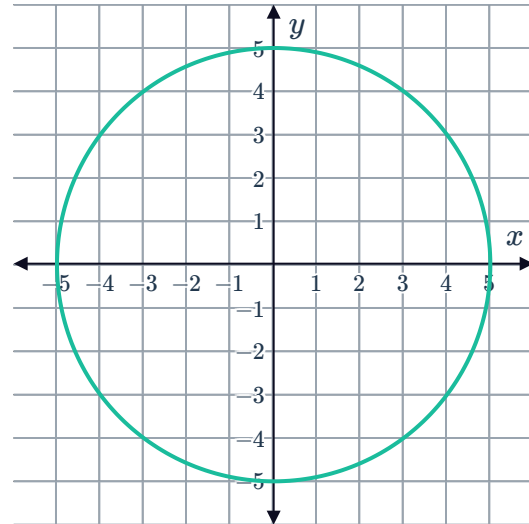


Graphs of circles



- 1 Consider the given graph of the circle.
Complete the statement:
Every point on the circle is exactly 5 units
away from the point (0, 0).



- 2 Determine whether the following points are inside, outside, or on the circle with equation $x^2 + y^2 = 25$:

a $(-3, 2)$

b $(4, 3)$

c $(1, 6)$

d $(-5, 0)$

- 3 For each of the following equations of circles:

i Find the radius.

ii Sketch the graph of the circle.

a $x^2 + y^2 = 16$

b $x^2 + y^2 = 36$

c $x^2 + y^2 = 25$

- 4 Sketch the graph of the circle given by $x^2 + y^2 = 3$.

- 5 For each of the following equations:

i Find the center of the circle.

ii Find the exact radius of the circle.

a $(x + 3.1)^2 + \left(y - \frac{3}{2}\right)^2 = \frac{9}{16}$

b $(x - 6)^2 + (y - 6)^2 = 12$

c $(x - 0.7)^2 + (y + 4.4)^2 = 3$

d $(x - 8)^2 + (y + 1)^2 = 13$

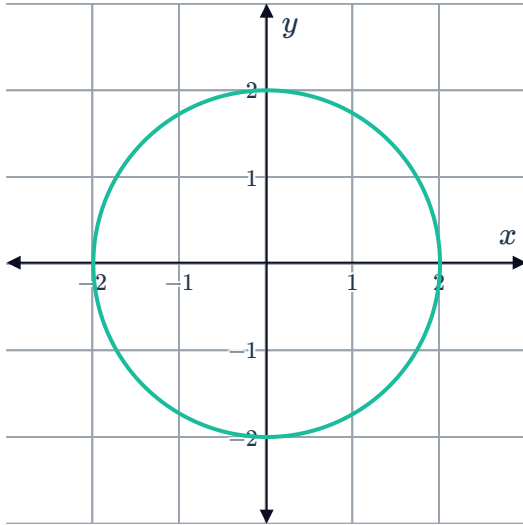
6 For each of the following circles, state:



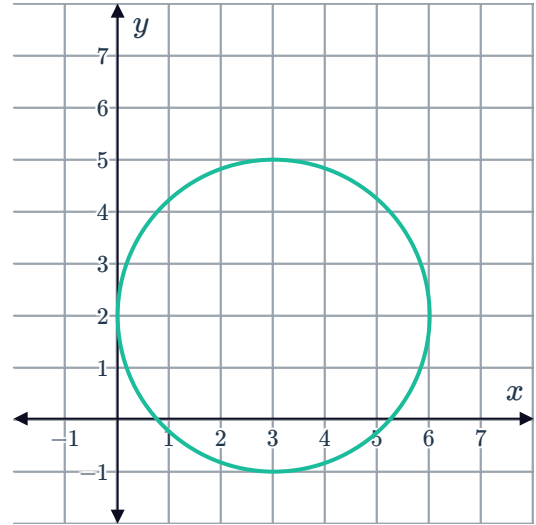
i The coordinates of the center.

ii The radius.

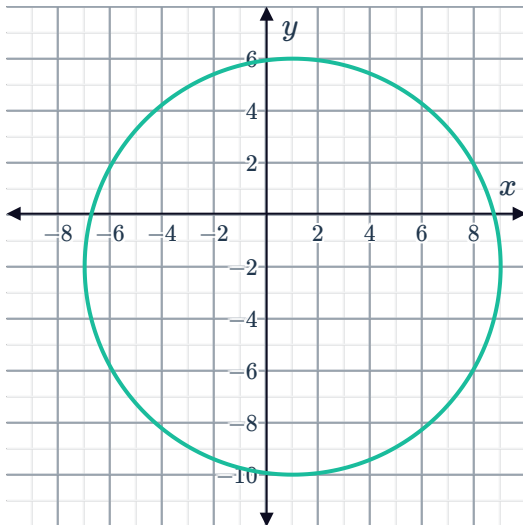
a



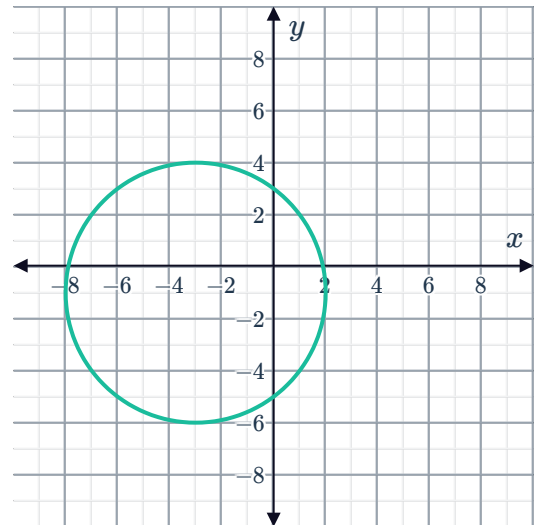
b



c



d



7 Sketch the graph of each of the following circles:

a $(x - 2)^2 + y^2 = 9$

b $(x + 3)^2 + y^2 = 16$

c $x^2 + (y + 3)^2 = 6^2$

8 For each of the following equations:



- i Find the center of the circle.
- ii Find the radius of the circle.
- iii Sketch the graph of the circle.

a $3x^2 + 3y^2 - 48 = 0$

b $25x^2 + 25y^2 = 400$

c $(x + 3)^2 + (y + 4)^2 = 16$

d $(x + 4)^2 + (y - 2)^2 = 25$

e $(x + 1)^2 + (y + 4)^2 = 25$

f $(x - 4)^2 + (y + 1)^2 = 36$

g $\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{9}{4}$

9 Consider the point $(-40, -9)$ that lies on the circle centered at the origin with a radius of 40 units:

- a Find the distance between $(-40, -9)$ and the origin.
- b Does the point $(-40, -9)$ lie on the circle?

10 Consider a circle with center O at $(0, 0)$ that passes through the point $A(-7, 6)$. A diameter is drawn from point A to point B on the circle.
State the coordinates of point B .

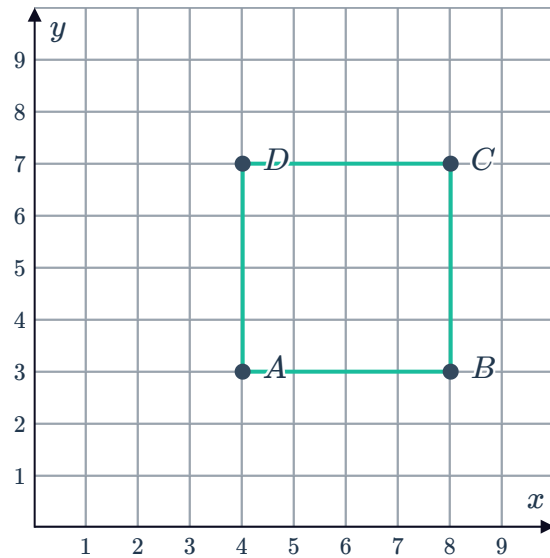
11 Consider the points $A(1, -4)$ and $B(9, 2)$ and a circle with center $C(5, -1)$:

- a Find the distance between points A and C .
- b Find the distance between points B and C .
- c State the coordinates of the midpoint of A and B .
- d Do points A and B lie at opposite ends of the circle?

- 12 Consider a circle drawn to fit inside square $ABCD$:



- State the coordinates of the center of the circle.
- Find the radius of the circle.



Equations of circles

- 13 State whether the following are equations of circles:

- | | | | |
|-------------------|---------------------|-------------------|----------------------|
| a $(x + y)^2 = 4$ | b $4x^2 + 5y^2 = 4$ | c $y^2 = 2 + x^2$ | d $4x^2 + 4y^2 = 16$ |
| e $x^2 + y^2 = 4$ | f $y^2 = 2 - x^2$ | | |

- 14 Consider the circle with center $(0, 0)$ that passes through the point $(5, -10)$.

- Find the radius of the circle.
- Find the equation of the circle.

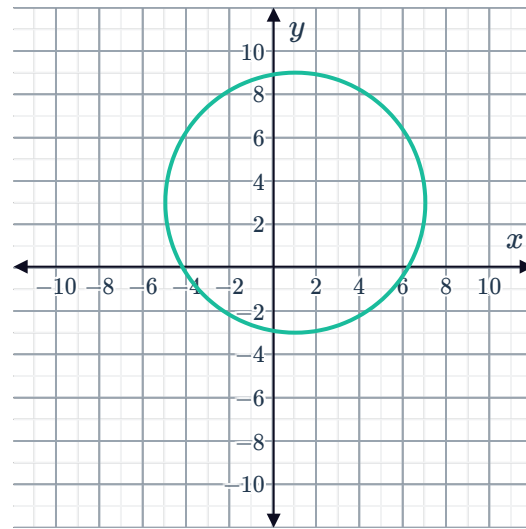
- 15 A circle with center $(0, 0)$ has an x -intercept at $(6, 0)$.

- Find the equation of the circle.
- Find the exact area inside the circle.

- 16 Select the equation of the circle whose center is closer to the origin:

A $x^2 + (y + 6)^2 = r^2$ **B** $x^2 + (y - 2)^2 = r^2$

17 Consider the circle on the graph:



- a State the coordinates of the center of the circle.
- b Find the radius of the circle.
- c State the equation of the circle in standard form.

18 For each of the following circles:

- i Sketch the graph.
 - ii Find the equation of the circle.
- a Circle with center at $(3, 3)$ and a radius of 6 units.
 - b Circle with center at $(0, 6)$ and a radius of 2 units.
 - c Circle with center at $(2, -6)$ and a radius of 3 units.
 - d A circle with center at the origin and a radius of 4 units.
 - e A circle with center at the origin and a radius of $\sqrt{3}$ units.
 - f A circle has its center at the origin and a radius of $\sqrt{18}$ units.

19 Consider the circle with center $(-2, 3)$ that passes through the point $(3, -4)$.

- a Find the radius of the circle.
- b State the equation of the circle in standard form.

20 For each of the following equations:

- i State the equation of the circle in standard form.
- ii State the coordinates of the center of the circle.
- iii Find the radius of the circle.

a $x^2 - 8x + y^2 - 20 = 0$

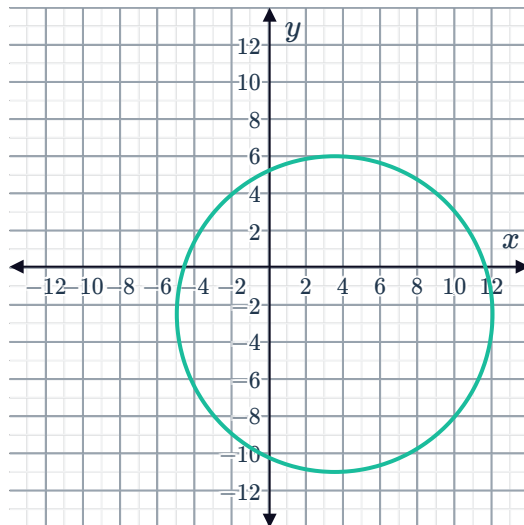
b $x^2 + 4x + y^2 + 6y - 3 = 0$

c $x^2 - 2x + y^2 + 8y - 19 = 0$

d $x^2 + 8x + y^2 - 6y + 24 = 0$

21 Consider the equation of a circle $x^2 + y^2 - 6x + \frac{27}{2} = 0$.

- State the equation of the circle in standard form.
- State the coordinates of the center of the circle.
- Find the radius of the circle.
- The circle that has been graphed has the same center as the circle with equation $x^2 + y^2 - 7x + 5y + \frac{27}{2} = 0$, but not the same radius. Would the circle with equation $x^2 + y^2 - 7x + 5y + \frac{27}{2} = 0$ lie inside or outside the given circle?



22 For each of the following equations:

- State the equation of the circle in standard form .
- State the coordinates of the center of the circle.
- Find the radius of the circle.
- Sketch the graph.

a $x^2 + y^2 + 2x + 4y = 11$

b $x^2 + 4x + y^2 - 10y = 52$

c $x^2 - 2x + y^2 - 16y - 16 = 0$

d $4x^2 + 4y^2 + 36x - 12y + 41 = 0$

23 Consider the equation of a circle given by $y^2 + 2y + 8 = 12x - x^2 + 7$.

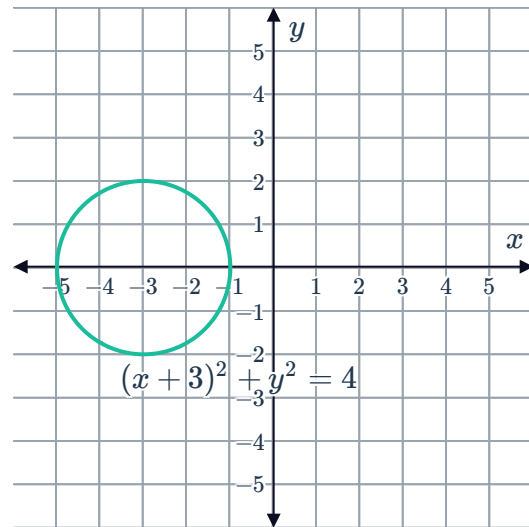
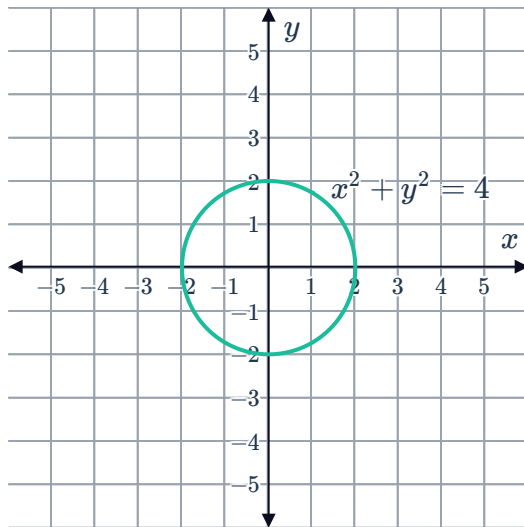
- State the equation of the circle in standard form.
- State the coordinates of the center of this circle.
- Find the radius of this circle.
- Sketch the graph of this circle.

24 The endpoints of the diameter of a circle are  $(-5, -3)$ and $(-15, -15)$.

- a** State the coordinates of the center of the circle.
- b** Find the exact radius of the circle.
- c** State the equation of the circle which has $(-5, -3)$ and $(-15, -15)$ as endpoints of its diameter.

Translations of circles

25 Consider the following circles:



Describe the translation from $x^2 + y^2 = 4$ to $(x + 3)^2 + y^2 = 4$.

26 For each of the following translations, state the equation of the resulting circle:

- a** A circle of radius 5, centered at the origin, is translated 2 units downwards.
- b** A circle of radius 7, centered at the origin, is translated 3 units vertically upwards.
- c** A circle of radius 3 units, centered at the origin, is translated 4 units to the right.
- d** A circle of radius 2 units, centered at the origin, is translated 5 units to the left.



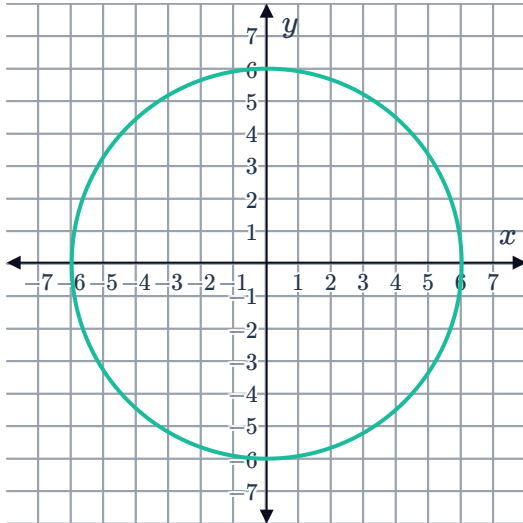
Domain and range of circles

27 For each of the following circles, state the following in interval notation:

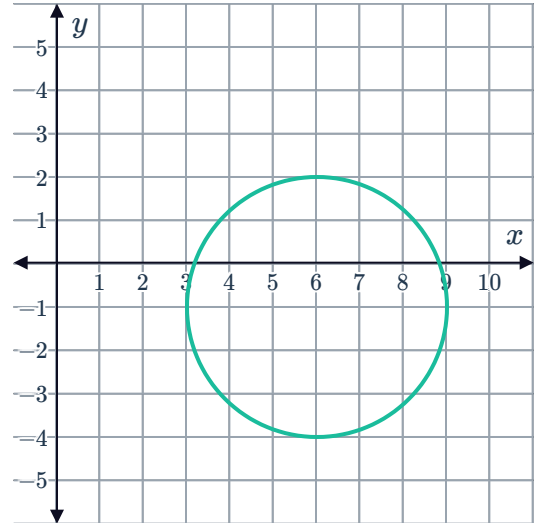
i The domain

ii The range

a



b



28 A circle has a radius of 1.5 and a center at $(-12, -13)$.

- a State the domain of the graph in interval notation.
- b State the range of the graph in interval notation.
- c Is the point $(-13, -22)$ inside or outside of the circle?

29 For each of the following equations:

- i Sketch the graph described by the equation.
- ii State the domain of the graph in interval notation.
- iii State the range of the graph in interval notation.

a $x^2 + y^2 = 36$

b $x^2 + (y - 4)^2 = 4$

c $(x + 5)^2 + (y + 3)^2 = 16$

30 A circle has a domain of $[2, 10]$ and a range of $[-4, 4]$.

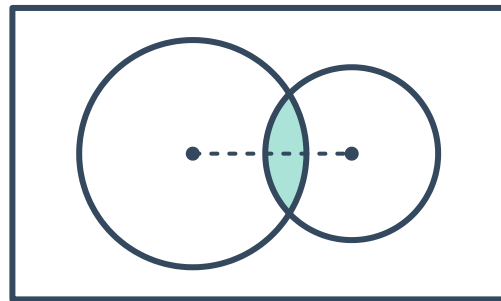
- a Sketch the graph of the circle.
- b State the equation for the circle.



Applications

- 31** Naval ship A picks up the signal of another vessel 45 km away. Ship A can also see that there is a partner naval ship, ship B , 30 km south and 40 km west of it.
- Using $(0, 0)$ as the position of naval ship A , determine the distance between naval ships A and B .
 - Could the signal that was picked up be coming from naval ship B ?
- 32** Three receiving stations are located on a coordinate plane, where each unit is one mile, at the points $(-3, -2)$, $(-1, 2)$ and $(3, 1)$. The epicenter of an earthquake is determined to be 2 miles, 4 miles and 5 miles from these three points respectively.
- Determine the coordinates of the epicenter of the earthquake.
 - The earthquake can be felt 10 miles away. Determine if the town located at $(8, -7)$ would feel the earthquake.

- 33** Amy needs to use her school's laser cutter to make a cork component for her design project. Using technology to sketch the component, she knows that the intersection of two different sized circles inscribed on a piece of cork gives the shape that she wants.



Two such circles have the equations $(x - 12)^2 + (y - 8)^2 = 49$ and $(x - 20)^2 + (y - 8)^2 = 36$, where each unit is 1 cm and $(0, 0)$ is the bottom left corner of the given piece of cork.

- State the domain and range of the leftmost circle in interval notation.
- State the domain and range of the rightmost circle in interval notation.
- What is the width of the component that Amy wants to make?
- Find the least area of material that could be used to make Amy's shape, considering the following constraints:
 - The laser cutter will only accept rectangular pieces of material.
 - The height of the intersection is 10.17 cm.
 - There also needs to be a gap of at least 1 cm between any cut and the edge of the piece.
- If the bottom left corner of the rectangular piece in part (d) is now $(0, 0)$, state the new



- 34** A hamster wheel has a radius of 12 cm. The clearance between the wheel and the ground is 5 cm. The origin is directly below the center of the wheel. Use the coordinate system to write the equation of the circular hamster wheel.
- 35** A rectangular coordinate system with coordinates in kilometers is placed with the origin at the center of a city. Beth lives in a unit located 5.3 km west and 5.4 km south of the city center. She is researching malt shops in her area and finds that there is one estimated to be 30 km from her unit. Write the standard form of the equation for the set of points that could be the location of the malt shop.